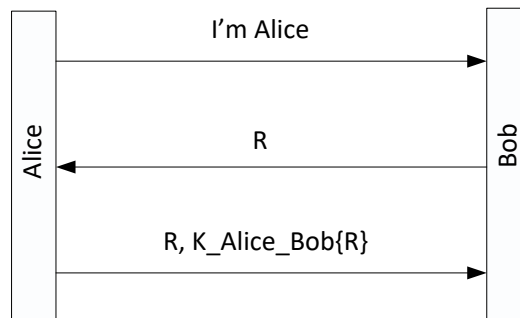


Homework 4 (100 pts)

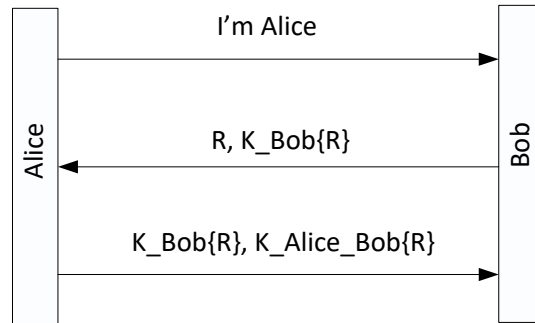
Rule: Finish all of the following on your own.

Think more *carefully* when you analyze an authentication protocol design.

1. Compute the following with DETAILED MATHEMATICAL STEPS. (Hints: Use Fermat Theorem, Euler Theorem, properties of totient functions, etc)
 - (1) $1234^{16} \bmod 17$
 - (2) $54^{51} \bmod 17$
 - (3) $\phi(51)$
 - (4) $\gcd(33, 121)$
 - (5) $2^{-1} \bmod 17$ (i.e., multiplicative inverse of 2 mod 17)
 - (6) $\text{ind}_{2,5}(4)$
 - (7) $\phi(8000)$
 - (8) $\phi(98803519)$
 - (9) $\phi(999866001989)$ for the graduate session only.
2. Describe the differences between digital signature and digital certificate.
3. Describe the differences between PKI and KDC.
4. Describe the differences between symmetric crypto and asymmetric crypto.
5. Explain the reflection attacks against symmetric key based authentication, and explain potential countermeasures.
6. Suppose we are using a three-message mutual authentication protocol, and Alice initiates contact with Bob. Suppose we wish Bob to be a stateless server, and therefore it is inconvenient to require him to remember the challenge he sent to Alice. Let's modify the exchange so that Alice sends the challenge back to Bob, along with the encrypted challenge. So the protocol is shown as follows. Is this protocol secure? Why?

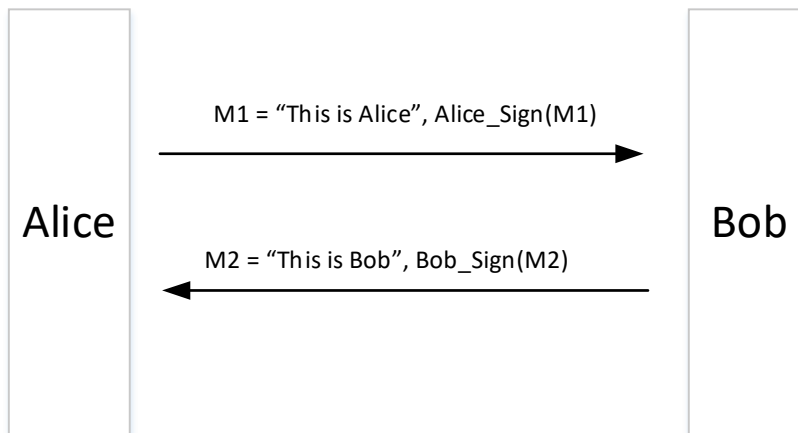


7. Let's modify the protocol from the previous problem so that Bob (still stateless) sends both a challenge, and a challenge encrypted with a key K_{Bob} that only he knows.

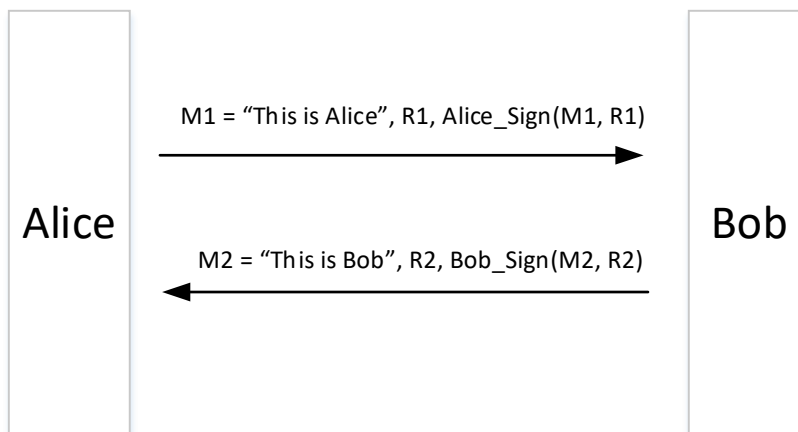


Is this protocol secure? Why?

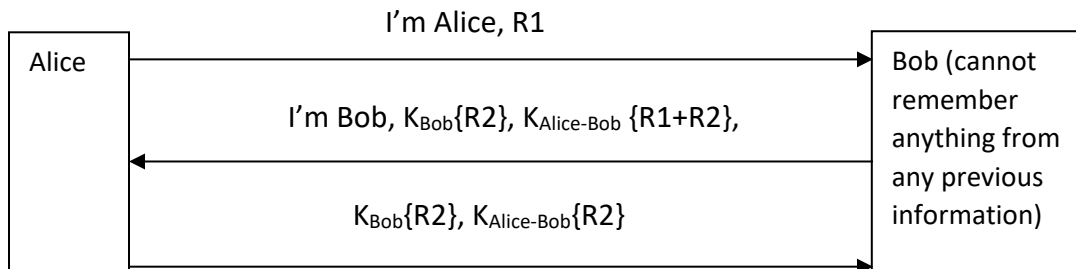
8. Suppose Alice and Bob know each other's public key, and they use the following protocol to achieve mutual authentication: Alice signs the message $M1 = \text{"This is Alice"}$ and Bob signs the message $M2 = \text{"This is Bob"}$. Explain the vulnerability.



9. Now, suppose Alice and Bob use random numbers to enhance security. Alice signs the combination of message $M1 = \text{"This is Alice"}$ and a random number $R1$, then send $M1$, $R1$, and digital signature to Bob; and Bob does the similar thing. Is this secure?



10. An even trickier question: In a three-message authentication protocol, Alice initiates contact with Bob. Bob is a stateless server, and thus it is inconvenient for him to remember the challenges sent to Alice. Assume that Alice and Bob share a key $K_{\text{Alice-Bob}}$ and Bob also has a key K_{Bob} only known to himself. The protocol works as follows, where $R1$ and $R2$ are random numbers generated by Alice and Bob, respectively.



- Does this protocol provide mutual authentication? If yes, why? If no, who cannot authenticate whom and why?
 - Could a third person, Trudy, impersonate Alice? Describe a possible attack scenario. If it is not possible, clearly state so and give reasons.
 - Could Trudy impersonate Bob? Describe a possible attack scenario. If it is not possible, clearly state so and give reasons.
11. What is the TCP/IP protocol? Specify what is the purpose of this protocol, and what kind of information it provides.
12. What is DNS, and how can ISP know the domain information that you want to visit?
13. What is the difference between using proxy server and the Tor?
14. Why proof of work is necessary to build a blockchain? Suppose if we remove the proof of work and design a simple rule: the first node in the network that finishes computing a new block always wins and this new block will be immediately broadcast to the network and all nodes must add this block to the chain. How will this simple (yet flawed) design impact the security?
15. (10 extra credit points for both graduates and undergraduates) The expanded Needham-Schroeder Protocol can be shortened to a 6-message protocol without loss of security by merely removing the 7th message. Why is this true? (Hint: the purpose of the 7th message is to prove to Bob that he is talking to Alice, but he already knows that. Why?)